

PARTS LIST

Semiconductors:

IC1	- 7805, 5V voltage regulator
IC2	- AT89C52 MCU
T1	- BC557 pnp transistor
D1-D3	- 1N4007 rectifier diode
LED1-LED2	- 5mm LED

Resistors (all 1/4-watt, $\pm 5\%$ carbon):

R1, R2	- 2.2-kilo-ohm
R3	- 10-kilo-ohm
R4-R17	- 1-kilo-ohm
R18	- 5.6-kilo-ohm
RNW1	- 10-kilo-ohm resistor network

Capacitors:

C1	- 470 μ F, 25V electrolytic
C2	- 10 μ F, 16V electrolytic
C3, C4	- 10nF ceramic disc
C5	- 100 μ F, 16V electrolytic
C6, C7	- 27pF ceramic disc

Miscellaneous:

TWS1, TWS2	- Thumbwheel switch (BCD type)
DIS1, DIS2	- LTS542 common-anode 7-segment display
S1	- On/off switch
RL1	- 5V single-changeover relay
XTAL1	- 12MHz crystal
LAMP1, LAMP2	- 60W, 230V AC bulb
X1	- 230V AC primary to 7.5V-0-7.5V, 500mA secondary transformer
	- 3-pin Berg female connector
	- 5-pin Berg male and female connector
	- 40-pin IC base



Fig. 5: Author's prototype of industrial timer

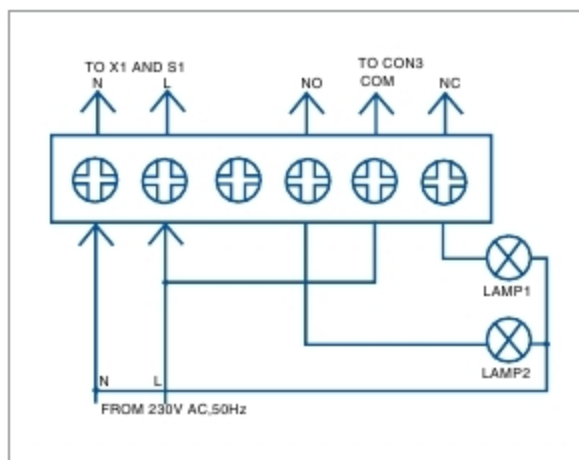


Fig. 6: Lamps connections

cabinet. Set TWS1 and TWS2 to any desired number. The author's prototype of the industrial timer is shown in Fig. 5.

For testing, connect 230V AC to L (line) and N (neutral) and two 60W bulbs for Lamp1 and Lamp2, as shown in Fig. 6. When power is switched on by closing S1, Lamp1 glows and Lamp2 goes in off condition.

The timer starts counting from 00 in the displays, and at the same time LED1 starts blinking. When counting reaches the set value, the timer stops. Then, Lamp1 and Lamp2 conditions get toggled.

After successful testing with lamps, disconnect the lamps. Now you can connect any electrical appliance to CON3 to operate it at a particular set time. **EFY**



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